MEA OCPP 1.6 Compliance Test Report

Section 11: Performance — Reconnect Time

Vector vSECC.single Board vs MEA CSMS (rddQC4000001)

MEA-V2G Project — Automated Test

2026-06-09 15:50

Contents

1 Test Configuration

Device Under Test	Vector vSECC.single Board
Device IP	192.168.1.166
OCPP Version	1.6 (JSON)
CP ID	rddQC4000001
CSMS	wss://ocpp.measandbox.com:2930
Connection	Direct (no proxy)
Test PC	192.168.111.185 (alias 192.168.1.200/24 on enp3s0)
Boot Wait Timeout	90 s per reconnect attempt
PASS Threshold	< 90 s
WARN Threshold	90–180 s
Repeat Count (11.2)	3 measurements
Delay Between Runs	5 s
Test Date	2026-06-09 08:44:44 UTC
Results file	tex/vsecc_section11_results.json

Measurement Method

For each reconnect timing:

1. The MQTT `_ocpp_up` event is cleared.
2. A **Hard Reset** command is issued via the MEA REST API (`POST /EV/remote/reset`).
3. The start time is recorded immediately after the reset call returns.
4. The test polls for two parallel reconnect signals:
   - MQTT topic `vsecc/ocpp_connection_status = connected`
   - MEA REST `POST /EV/cmd/chargepoint/getConfiguration` returning HTTP 200
5. The elapsed time is recorded when either signal is observed.

The timing includes the vSECC boot time, TLS handshake, and OCPP BootNotification / BootNotification-Accepted exchange.

2 Section 11 Results

Summary: 2 PASS 0 FAIL 0 WARN 0 SKIP (2 total)

Item	Test Description	Detail / Evidence	Result
11.1 Reconnect Time — Single Measurement			
11.1	Reconnect Time after Hard Reset	`Reconnected in 0.3 s (< 90 s threshold)` *time_ms=257*	PASS
11.2 Average Reconnect Time — Repeated Measurements			

11.2	Average Reconnect Time (3 measurements)	`avg=0.3s min=0.2s max=0.3s [0.3, 0.2, 0.3]`
		`run1=260 ms`
		`run2=239 ms`
		`run3=258 ms`
			PASS

3 Reconnect Time Measurements

Item 11.1 — Single measurement

A single Hard Reset is issued and the time to reconnect is measured. The result is classified as:

- **PASS** if reconnect time < 90 s

- **WARN** if reconnect time is 90–180 s (marginal)
- **FAIL** if no reconnect within 180 s

The timing captures the full reboot cycle: OS startup, V2G stack initialization, WebSocket TLS connection to the CSMS, and OCPP BootNotification exchange.

Item 11.2 — Repeated measurements

The reconnect measurement is repeated 3 times with 5 s between runs. Individual run timings are shown in the detail column above. The average, minimum, and maximum times are reported in the detail field. The item is classified as PASS if the *average* reconnect time < 90 s.

Factors affecting reconnect time

- **Boot time:** Linux kernel + systemd startup on the vSECC hardware.
- **TLS handshake:** Certificate validation and TLS 1.2/1.3 negotiation with `wss://ocpp.measandbox.com:2930`.
- **OCPP boot sequence:** BootNotification + BootNotificationAccepted + initial StatusNotification messages.
- **Network latency:** Round-trip time to the MEA sandbox CSMS in Thailand.

4 Notes

Detection method

The test uses a dual detection strategy:

1. **MQTT:** The vSECC publishes `connected` to `vsecc/ocpp_connection_status` when the OCPP WebSocket is established. This is the most precise signal.
2. **REST probe:** A `GetConfiguration` call is made to the MEA CSMS every second. HTTP 200 indicates the CSMS can forward commands to the charger, confirming connectivity.

Whichever signal arrives first determines the recorded reconnect time.